

FIREWALL BASICS THEORY

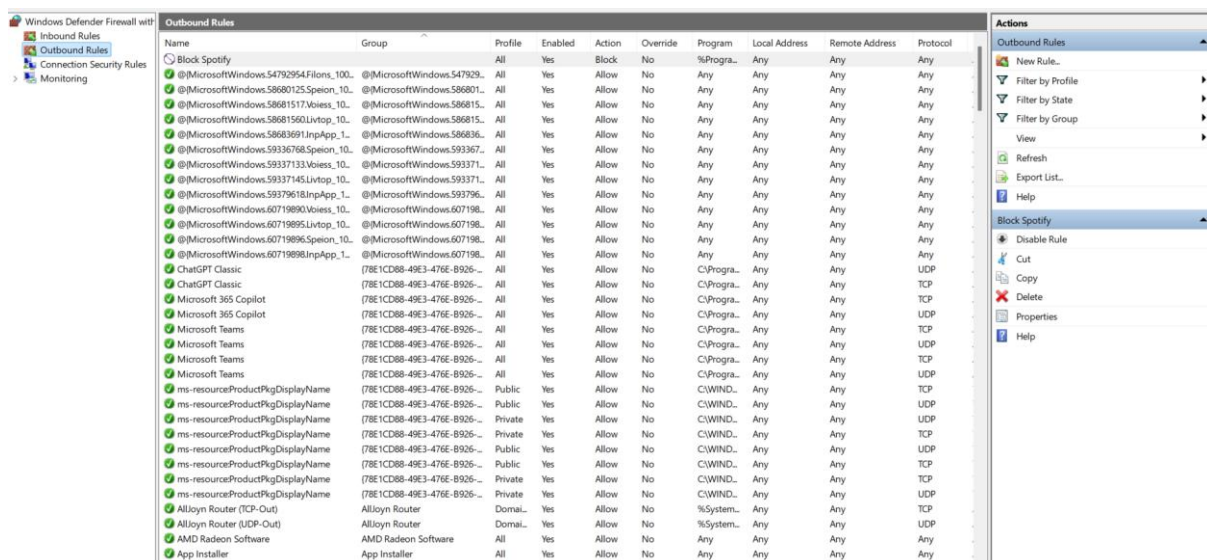
1. Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.



Steps

1. Press Windows + R, type wf.msc, and press Enter to open Windows Defender Firewall with Advanced Security.
2. In the left panel, click Outbound Rules.
3. In the right panel, click New Rule....
4. In the New Outbound Rule Wizard, select Program and click Next.
5. Select This program path and click Browse.
6. Locate and select the Spotify.exe file.
(If Spotify.exe is not found, open Spotify → press Ctrl + Shift + Esc → right-click Spotify in Task Manager → Open file location, then select Spotify.exe.)
7. Click Next.
8. Select Block the connection and click Next.
9. Keep Domain, Private, and Public profiles selected, then click Next.
10. In the Name field, type Block Spotify.
11. Click Finish to create the outbound firewall rule.
12. Open the Spotify desktop application and try to play a song. The application will not be able to access the internet because the outbound traffic has been blocked.

Result



2. Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).



Steps

1. Press Windows + R, type wf.msc, and press Enter.
2. The Windows Defender Firewall with Advanced Security window will open.
3. In the left pane, click Inbound Rules.
4. In the right pane, click New Rule....
5. Select Program and click Next.
6. Select This program path and click Browse.
7. Browse to the Google Chrome installation folder and select chrome.exe.

○ Default path:

C:\Program Files\Google\Chrome\Application\chrome.exe

or

C:\Program Files (x86)\Google\Chrome\Application\chrome.exe

8. Click Next.
9. Select Allow the connection and click Next.
10. Leave Domain, Private, and Public checked, then click Next.
11. Enter the rule name as Allow Google Chrome and click Finish.
12. Again, click New Rule....
13. Select Port and click Next.
14. Select TCP.
15. Choose Specific local ports and enter:
443
16. Click Next.
17. Select Allow the connection and click Next.
18. Leave Domain, Private, and Public selected, then click Next.
19. Enter the rule name as HTTPS Port 443.
20. Click Finish.

Inbound Rules										Actions	
Name	Group	Profile	Enabled	Action	Override	Program	Local Address	Remote Address	Protocol		
HTTPS Port 443		All	Yes	Allow	No	Any	Any	Any	TCP	Inbound Rules	
Allow Chrome HTTPS		All	Yes	Allow	No	%Progra...	Any	Any	Any	New Rule...	
AnyDesk		Private	Yes	Allow	No	C:\Progra...	Any	Any	UDP	Filter by Profile	
AnyDesk		Domain	Yes	Allow	No	C:\Progra...	Any	Any	TCP	Filter by State	
AnyDesk		Public	Yes	Allow	No	C:\Progra...	Any	Any	UDP	Filter by Group	
AnyDesk		Public	Yes	Allow	No	C:\Progra...	Any	Any	TCP	View	
AnyDesk		Domain	Yes	Allow	No	C:\Progra...	Any	Any	UDP	Refresh	
AnyDesk		Private	Yes	Allow	No	C:\Progra...	Any	Any	TCP	Export List...	
FTP SERVER		All	Yes	Allow	No	Any	Any	Any	TCP	Help	
ChatGPT Classic	(78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progra...	Any	Any	UDP	Selected Items	
ChatGPT Classic	(78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progra...	Any	Any	TCP	Disable Rule	
Microsoft 365 Copilot	(78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progra...	Any	Any	TCP	Cut	
Microsoft 365 Copilot	(78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progra...	Any	Any	UDP	Copy	
Microsoft Teams	(78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progra...	Any	Any	TCP	Delete	
Microsoft Teams	(78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progra...	Any	Any	UDP	Help	
Microsoft Teams	(78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progra...	Any	Any	TCP		
Microsoft Teams	(78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progra...	Any	Any	UDP		
Microsoft Teams (personal)	(78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progra...	Any	Any	TCP		
Microsoft Teams (personal)	(78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progra...	Any	Any	TCP		
ms-resourceAppTitle	(78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progra...	Any	Any	TCP		
ms-resourceAppTitle	(78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progra...	Any	Any	UDP		

3. Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.
Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.



Steps

1. Press Windows + R, type wf.msc, and press Enter to open Windows Defender Firewall with Advanced Security.
2. In the left pane, click Inbound Rules.
3. In the right pane, click New Rule....
4. Select Custom and click Next.
5. Leave All programs selected and click Next.
6. Leave the default Protocol and Ports settings unchanged and click Next.
7. On the Scope page:
 - Under Which remote IP addresses does this rule apply to?, select These IP addresses.
 - Click Add....
 - Enter the IP address: 123.45.67.89
 - Click OK, then click Next.
8. Select Block the connection and click Next.
9. Keep Domain, Private, and Public profiles selected, then click Next.
10. Enter the rule name: Block IP 123.45.67.89
11. Click Finish to create the rule.
12. Verify that the new rule appears in the Inbound Rules list.

4. Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.



➤ Feature	Hardware Firewall	Software Firewall
Installation	Installed as a separate physical device between the network and the Internet.	Installed as software on an individual computer or laptop.
Protection	Protects all devices connected to the network.	Protects only the computer on which it is installed.
Performance	Handles high network traffic with minimal impact on user devices.	Uses the computer's CPU and memory, which may slightly affect performance.

Use Case of Hardware Firewall

Large organizations, schools, and offices use hardware firewalls to secure the entire network from unauthorized access and cyber threats.

Use Case of Software Firewall

Windows Defender Firewall is commonly used on personal computers and laptops to monitor and control incoming and outgoing network traffic for individual devices.